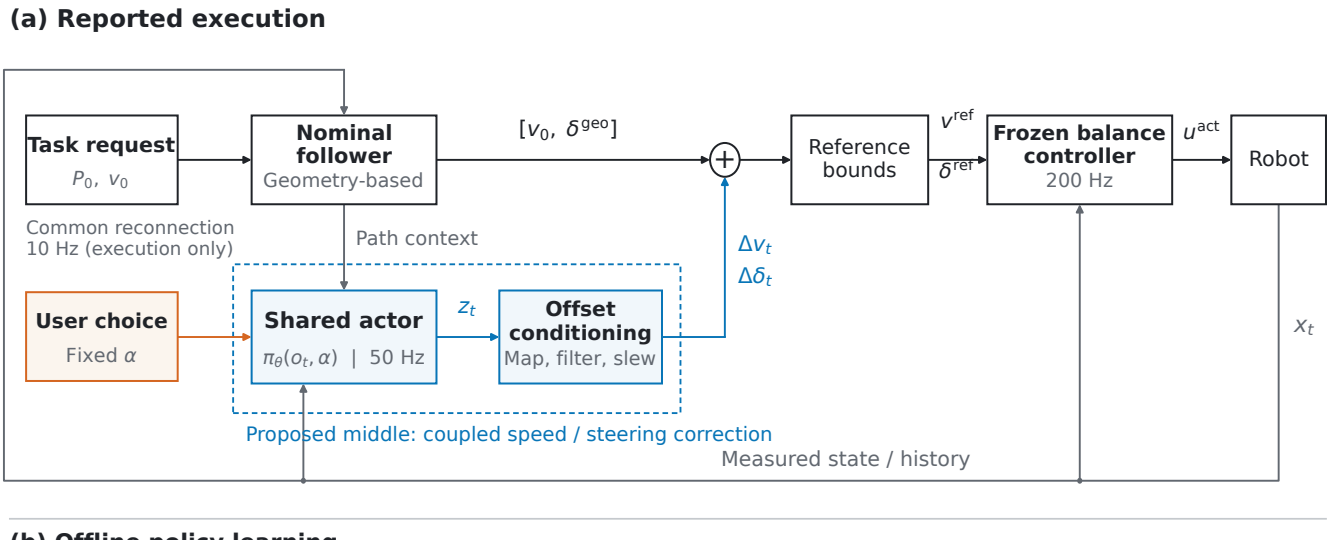
(a) Reported execution
Task request
$P_0,\ v_0$
Nominal follower
Geometry-based
$[v_0,\ \delta^{geo}]$
Reference bounds
v^{ref}
δ^{ref}
Frozen balance controller
200 Hz
u^{act}
Robot
Common reconnection
10 Hz (execution only)
Path context
Δv_t
$\Delta \delta_t$
x_t
User choice
Fixed α
Shared actor
$\pi_\theta(o_t, \alpha)$ | 50 Hz
z_t
Offset conditioning
Map, filter, slew
Proposed middle: coupled speed / steering correction
Measured state / history
(b) Offline policy learning
Fixed-path rollouts
Frozen training lower
PPO actor-critic + parent anchor
Preference-weighted reward r_α
Freeze shared actor
Same θ for all tested preferences